

Monoidal extensivity and the comodule fibration

Part I

Matthew Di Meglio
m.dimeglio@ed.ac.uk



THE UNIVERSITY
of EDINBURGH

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- 1 Indexed symmetric monoidal categories and symmetric monoidal fibrations
- 2 Comonoids and comodules
- 3 Monoidal extensivity

An *indexed symmetric monoidal category* is a pseudofunctor

$$V: \mathbf{S}^{\text{op}} \rightarrow \mathbf{SymMonCat}$$

where

- $\mathbf{S}$ is a cartesian monoidal category, and
- $\mathbf{SymMonCat}$ is the 2-category of symmetric monoidal categories, strong symmetric monoidal functors and monoidal natural transformations.

Notation:

- $(V^J, \otimes_J, I_J)$ is the image of J ,
- $V^K \xrightarrow{\Delta_r} V^J$ is the image of $J \xrightarrow{r} K$,
- $(V, \otimes, I)$ is the image of 1 ,
- $V \xrightarrow{\Delta_J} V^J$ is the image of $J \xrightarrow{e_J} 1$.

An *indexed symmetric monoidal functor*

$$(F, G): (\mathbf{S}^{\text{op}} \xrightarrow{V} \mathbf{SymMonCat}) \rightarrow (\mathbf{T}^{\text{op}} \xrightarrow{W} \mathbf{SymMonCat})$$

consists of

- a strong monoidal functor $F: \mathbf{S} \rightarrow \mathbf{T}$, and
- a pseudonatural transformation $G: V \rightarrow W \circ F$.

Notation:

- $G^J: V^J \rightarrow W^{FJ}$ is the J -component of G

A *symmetric monoidal fibration* is a strict symmetric monoidal functor

$$P: \mathbf{A} \rightarrow \mathbf{S}$$

where

- $\mathbf{S}$ is cartesian monoidal,
- P is a fibration, and
- the monoidal product of two P -cartesian morphisms is itself P -cartesian.

Notation:

- $\boxtimes$ and I are the monoidal product and monoidal unit of $\mathbf{A}$,
- $\delta_r A: \Delta_r A \rightarrow A$ is a chosen P -cartesian lift of $r: J \rightarrow K$,
- $\delta_J A: \Delta_J A \rightarrow A$ is a chosen P -cartesian lift of $!: J \rightarrow 1$,

Handwritten red symbol resembling a stylized 'J' or 'L'.

A *morphism of symmetric monoidal fibrations*

$$(F, G): (\mathbf{A} \xrightarrow{P} \mathbf{S}) \rightarrow (\mathbf{B} \xrightarrow{Q} \mathbf{T})$$

is a pair of strong monoidal functors

$$F: \mathbf{S} \rightarrow \mathbf{T} \quad \text{and} \quad G: \mathbf{A} \rightarrow \mathbf{B}$$

such that

- $Q \circ G = F \circ P$,
- the identity natural transformation $Q \circ G \rightarrow F \circ P$ is monoidal, and
- F preserves cartesian morphisms.

Proposition (See Shulman, 2008)

*The 2-categories **IndSymMonCat** and **SymMonFib** are equivalent.*

Given $P: \mathbf{A} \rightarrow \mathbf{S}$, the J -fibre of P is symmetric monoidal with

$$I_J = \Delta_J I \quad \text{and} \quad X \otimes_J Y = \Delta_{d_J}(X \boxtimes Y)$$

Given $V: \mathbf{S}^{\text{op}} \rightarrow \mathbf{SymMonCat}$, the category $\int V$ is symmetric monoidal with

$$I = I_1 \quad \text{and} \quad X \boxtimes Y = \Delta_{p_1} X \otimes_{J \times K} \Delta_{p_2} Y$$

Notation:

- d_J is the diagonal map $J \rightarrow J \times J$
- p_1 and p_2 are the product projections $J \times K \rightarrow J$ and $J \times K \rightarrow K$

The **FinSet**-indexing of a symmetric monoidal category $\mathcal{V}$ is the indexed symmetric monoidal category

$$\mathbf{FinFam}_{\mathcal{V}}: \mathbf{FinSet}^{\mathrm{op}} \rightarrow \mathbf{SymMonCat},$$

where

- for each finite set J , the category $\mathbf{FinFam}_{\mathcal{V}}(J)$ has
 - *objects*: J -indexed families $(X_j)_{j \in J}$ of objects of $\mathcal{V}$
 - *morphisms* $(X_j)_{j \in J} \rightarrow (Y_j)_{j \in J}$: J -indexed families of morphisms $(f_j)_{j \in J}$ of $\mathcal{V}$ where $f_j: X_j \rightarrow Y_j$
 - *monoidal unit*: $(I)_{j \in J}$
 - *monoidal product*: $(X_j)_{j \in J} \otimes (Y_j)_{j \in J} = (X_j \otimes Y_j)_{j \in J}$
- for each function $r: J \rightarrow K$ between finite sets, $\Delta_r(Y_k)_{k \in K} = (Y_{rj})_{j \in J}$

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Let $\mathcal{V}$ be a symmetric monoidal category.

A (**cocommutative**) **comonoid** J is an object J equipped with morphisms

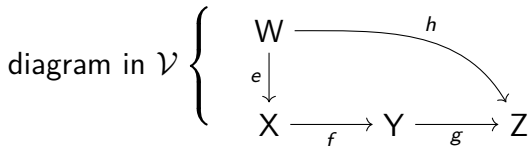
$$\begin{array}{ccc} d_J: J \rightarrow J \otimes J & & e_J: J \rightarrow I \\ \text{\textit{comultiplication}} & \text{and} & \text{\textit{counit}} \end{array}$$

that are coassociative, counital and cocommutative.

A **comonoid morphism** $r: J \rightarrow K$ is a morphism $r: J \rightarrow K$ that preserves the counit and comultiplication.

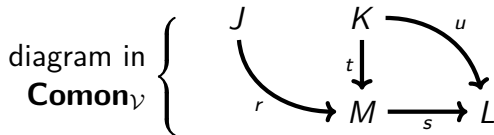
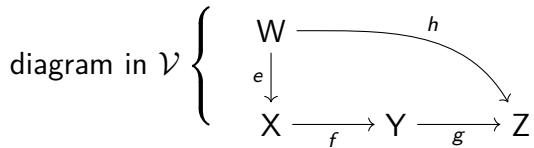
Let **Comon** $_{\mathcal{V}}$ be the category of comonoids and comonoid morphisms in $\mathcal{V}$.

A **coaction** of a comonoid J on an object X is a morphism $x: X \rightarrow J \otimes X$ that is coassociative and counital with respect to J .

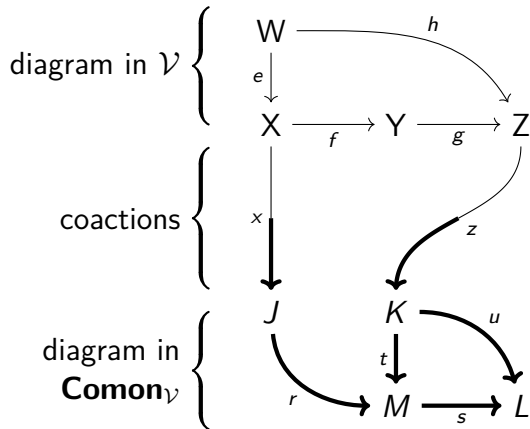


Build

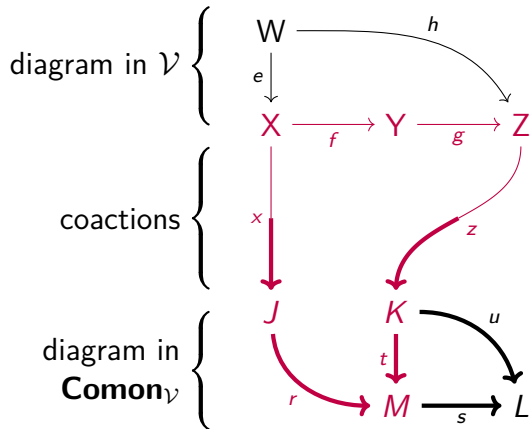
A **coaction** of a comonoid J on an object X is a morphism $x: X \rightarrow J \otimes X$ that is coassociative and counital with respect to J .



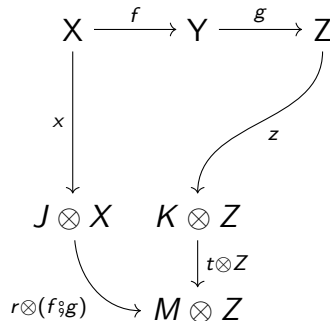
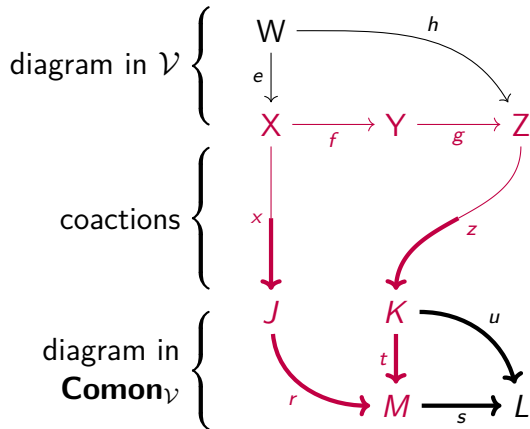
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Let $\mathbf{Comod}_{\mathcal{V}}$ be the category of comodules in $\mathcal{V}$. It has

- *objects*: coactions $X \xrightarrow{x} J$

- *morphisms* $X \xrightarrow{x} J$ to $Y \xrightarrow{y} K$: commuting squares

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ x \downarrow & & \downarrow y \\ J & \xrightarrow{r} & K \end{array}$$

is mon

It is a symmetric monoidal analogue of the arrow category $\mathcal{V}^2$ with

$$I = \begin{array}{c} I \\ \downarrow \ell_I^{-1} \\ I \end{array}$$

and

$$\begin{array}{ccc} X & Y & X \otimes Y \\ x \downarrow & \boxtimes & \downarrow y \\ J & K & J \otimes K \end{array} = \begin{array}{c} X \otimes Y \\ \downarrow x \otimes y \circ J \otimes s_{X,K} \otimes Y \\ J \otimes K \end{array}$$

Let $\text{Comod}_{\mathcal{V}}: \mathbf{Comod}_{\mathcal{V}} \rightarrow \mathbf{Comon}_{\mathcal{V}}$ be the functor given on morphisms by

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ x \downarrow & & \downarrow y \\ J & \xrightarrow{r} & K \end{array} \quad \mapsto \quad J \xrightarrow{r} K$$

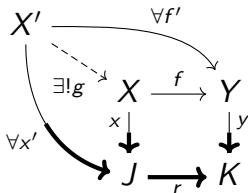
It is a strict symmetric monoidal analogue of the codomain functor $\mathcal{V}^2 \rightarrow \mathcal{V}$.

Proposition

A morphism $\begin{array}{ccc} X & \xrightarrow{f} & Y \\ x \downarrow & & \downarrow y \\ J & \xrightarrow{r} & K \end{array}$ is $\text{Comod}_{\mathcal{V}}$ -opcartesian if and only if f is an isomorphism.

It follows that $\text{Comod}_{\mathcal{V}}$ is an opfibration.

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ x \downarrow & & \downarrow y \\ J & \xrightarrow{r} & K \end{array}$$
 is a **coinduction square** and (x, f) is a **coinduction** of (y, r) if



Proposition

A morphism is $\text{Comod}_{\mathcal{V}}$ -cartesian if and only if it is a coinduction square.

Corollary

If $\mathcal{V}$ has all coinductions, then $\text{Comod}_{\mathcal{V}}$ is a symmetric monoidal fibration.

A symmetric monoidal category $\mathcal{V}$ has *nice coreflexive equalisers* if

- it has coreflexive equalisers, and
- for all objects X of $\mathcal{V}$, the functor $X \otimes -$ preserves coreflexive equalisers.

An indexed symmetric monoidal category $V: \mathbf{S}^{\text{op}} \rightarrow \mathbf{SymMonCat}$ has *nice coreflexive equalisers* if

- each category V^J has nice coreflexive equalisers, and
- each indexed diagonal functor Δ_r preserves coreflexive equalisers.

(or, equivalently, if V factors through $\mathbf{EqSymMonCat} \hookrightarrow \mathbf{SymMonCat}$)

A symmetric monoidal fibration $P: \mathbf{A} \rightarrow \mathbf{S}$ has *nice coreflexive equalisers* if

- P lifts equalisers of coreflexive pairs, and
- for all objects X of $\mathbf{A}$, lifted coreflexive equalisers are preserved by $X \otimes -$.

Proposition

A symmetric monoidal fibration has nice coreflexive equalisers if and only if the corresponding indexed symmetric monoidal category has nice coreflexive equalisers.

Proposition

*A symmetric monoidal category $\mathcal{V}$ has nice coreflexive equalisers if and only if the **Set**-indexing of $\mathcal{V}$ has nice coreflexive equalisers.*

Proposition

Let $\mathcal{V}$ be a symmetric monoidal category with nice coreflexive equalisers. Then then $\text{Comod}_{\mathcal{V}}$ is a Beck-Chevalley symmetric monoidal bifibration with nice coreflexive equalisers.

Let $\mathbf{Comod}_{\mathcal{V}}(J)$ be the J -fibre of $\mathbf{Comod}_{\mathcal{V}}$. It has

- *objects*: coactions $X \xrightarrow{x} J$
- *morphisms* $X \xrightarrow{x} J$ to $X' \xrightarrow{x'} J$: commuting triangles

$$\begin{array}{ccc} X & \xrightarrow{f} & X' \\ & \searrow x & \swarrow x' \\ & J & \end{array}$$

It is an analogue of the slice category $\mathcal{V}/J$.

If $\mathcal{V}$ has all coinductions, then $\mathbf{Comod}_{\mathcal{V}}(J)$ becomes symmetric monoidal via the equivalence $\mathbf{SymMonFib} \simeq \mathbf{IndSymMonCat}$.

Proposition

Let $P: \mathbf{A} \rightarrow \mathbf{S}$ be a symmetric monoidal fibration. Then

$$\mathbf{Comon}_P: \mathbf{Comon}_\mathbf{A} \rightarrow \mathbf{Comon}_\mathbf{S} \quad \text{and} \quad \mathbf{Comod}_P: \mathbf{Comod}_\mathbf{A} \rightarrow \mathbf{Comod}_\mathbf{S}$$

are both symmetric monoidal fibrations, and the forgetful morphisms

$$\mathbf{Comon}_P \rightarrow P \quad \text{and} \quad \mathbf{Comod}_P \rightarrow P \times \mathbf{Comon}_P$$

of symmetric monoidal fibrations both create cartesian and opcartesian lifts.

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An indexed symmetric monoidal category

$$V: \mathbf{S}^{\text{op}} \rightarrow \mathbf{SymMonCat}$$

is *monoidal extensive* if

- each indexed diagonal functor Δ_r has a left adjoint Σ_r , and
- for each $r: J \rightarrow K$ in $\mathbf{S}$ and each comonoid A in V^J , the canonical functor

$$\Sigma_r: \mathbf{Comod}_{V^J}(A) \rightarrow \mathbf{Comod}_{V^K}(\Sigma_r A)$$

is an equivalence of categories.

A symmetric monoidal fibration $P: \mathbf{A} \rightarrow \mathbf{S}$ is *monoidal extensive* if

- it is an opfibration,
- coinductions of vertical comodules along P -opcartesian comonoid morphisms exist and are vertical, and
- each commuting mixed square

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ x \downarrow & & \downarrow y \\ A & \xrightarrow{\sigma_r A} & \Sigma_r A \end{array}$$

in $\mathbf{A}$, where x and y are vertical comodules, is a coinduction square if and only if f is P -opcartesian.

Proposition

A symmetric monoidal fibration $P: \mathbf{A} \rightarrow \mathbf{S}$ is monoidal extensive if and only if the corresponding indexed symmetric monoidal category $V: \mathbf{S}^{\text{op}} \rightarrow \mathbf{SymMonCat}$ is monoidal extensive.

Lemma

$$\mathbf{Comon}_{VJ} \cong (\mathbf{Comon}_P)^J$$

$$\begin{array}{ccc} & A & \\ d'_A \swarrow & & \searrow d_A \\ A \otimes_J A & \xrightarrow{\delta_{d_J}(A \boxtimes A)} & A \boxtimes A \end{array}$$

$$\begin{array}{ccc} & A & \\ e'_A \swarrow & & \searrow e_A \\ I_J & \xrightarrow{\delta_{e_J} I} & I \end{array}$$

$\xrightarrow{P}$

$$\begin{array}{ccc} & J & \\ \parallel \swarrow & & \searrow d_J \\ J & \xrightarrow{d_J} & J \times J \end{array}$$

$\xrightarrow{P}$

$$\begin{array}{ccc} & J & \\ \parallel \swarrow & & \searrow e_J \\ J & \xrightarrow{e_J} & 1 \end{array}$$

Lemma

$$\mathbf{Comod}_{V^J}(A) \cong (\mathbf{Comod}_P)^J(A)$$

$$\begin{array}{ccc}
 & X & \\
 x' \swarrow & & \searrow x \\
 A \otimes_J X & \xrightarrow{\delta_{d_J}(A \boxtimes X)} & A \boxtimes X
 \end{array}$$

$$P \mapsto$$

$$\begin{array}{ccc}
 & J & \\
 \parallel \swarrow & & \searrow d_J \\
 J & \xrightarrow{d_J} & J \times J
 \end{array}$$

Lemma

The diagram

$$\begin{array}{ccc}
 \mathbf{Comod}_{\mathcal{V}^J}(A) & \xrightarrow{\Sigma_r} & \mathbf{Comod}_{\mathcal{V}^K}(\Sigma_r A) \\
 \cong \downarrow & & \downarrow \cong \\
 (\mathbf{Comod}_P)^J(A) & \xrightarrow{\Sigma_r} & (\mathbf{Comod}_P)^K(\Sigma_r A)
 \end{array}$$

commutes.

Hence $\mathcal{V}$ is monoidal extensive if and only if

- P is an opfibration, and
- for each $r: J \rightarrow K$ in $\mathbf{S}$ and each A in $(\mathbf{Comon}_P)^J$, the functor

$$(\mathbf{Comod}_P)^J(A) \xrightarrow{\Sigma_r} (\mathbf{Comod}_P)^K(\Sigma_r A)$$

is an equivalence of categories.

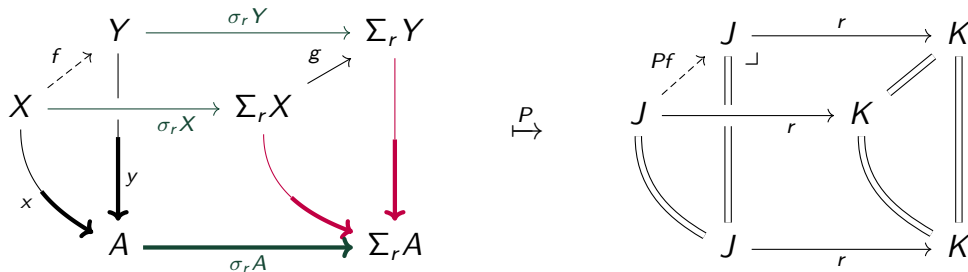
Suppose that $P: \mathbf{A} \rightarrow \mathbf{S}$ is monoidal extensive.

Let $r: J \rightarrow K$ in $\mathbf{S}$ and A in $(\mathbf{Comon}_P)^J$.

We must show that

$$(\mathbf{Comod}_P)^J(A) \xrightarrow{\Sigma_r} (\mathbf{Comod}_P)^K(\Sigma_r A)$$

is full, faithful and essentially surjective.



- The left-hand back face is a coinduction square by extensivity of P .
- Obtain unique f from the coinduction universal property.
- $Pf = \text{id}_J$ from the pullback universal property of the right-hand back face.
- $g = \Sigma_r f$ from opcartesian universal property of $\sigma_r X$.

$$\begin{array}{ccc}
 X & \xrightarrow{\quad f \quad} & Y \\
 \downarrow x & \lrcorner & \downarrow y \\
 A & \xrightarrow{\quad \sigma_r A \quad} & \Sigma_r A
 \end{array}
 \xrightarrow{P}
 \begin{array}{ccc}
 J & \xrightarrow{\quad r \quad} & K \\
 \parallel & \lrcorner & \parallel \\
 A & \xrightarrow{\quad r \quad} & K
 \end{array}$$

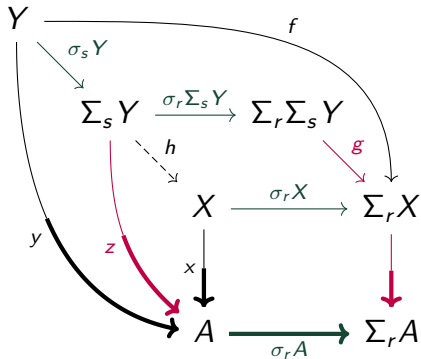
- Obtain coinduction lift (x, f) from extensivity of P .
- f is P -opcartesian as P is extensive.
- $(Y, y) = \Sigma_r(X, x)$ from universal property of **Comod** $_P$ -opcartesian morphism $(f, \sigma_r A)$.

Conversely, suppose that

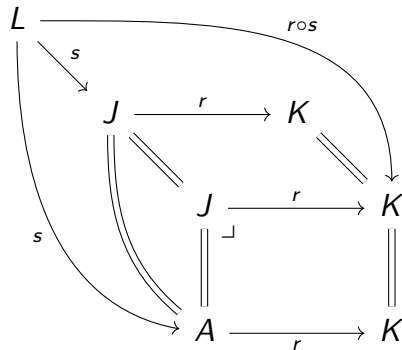
- P is an opfibration, and
- for each $r: J \rightarrow K$ in $\mathbf{S}$ and each A in $(\mathbf{Comon}_P)^J$, the functor

$$(\mathbf{Comod}_P)^J(A) \xrightarrow{\Sigma_r} (\mathbf{Comod}_P)^K(\Sigma_r A)$$

is an equivalence of categories.



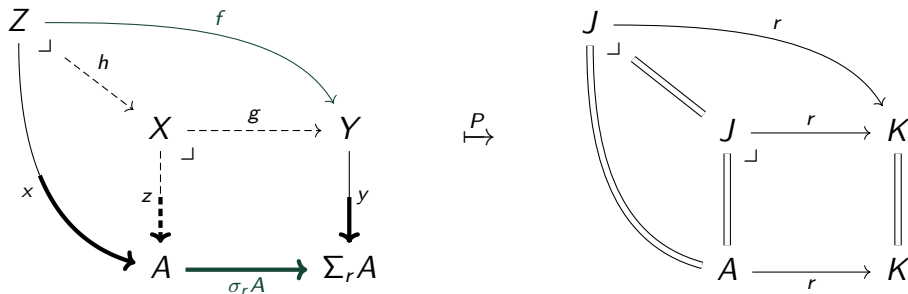
$\xrightarrow{P}$



Obtain unique h from fullness and faithfulness of Σ_r .

$$\begin{array}{ccc}
 X & \xrightarrow{\quad f \quad} & Y \\
 \downarrow x & & \downarrow y \\
 A & \xrightarrow{\quad \sigma_r A \quad} & \Sigma_r A
 \end{array}
 \quad \xrightarrow{P} \quad
 \begin{array}{ccc}
 J & \xrightarrow{\quad r \quad} & K \\
 \parallel \lrcorner & & \parallel \\
 A & \xrightarrow{\quad r \quad} & K
 \end{array}$$

- From essential surjectivity of Δ_r , obtain (x, f) where f is P -opcartesian.
- Left-hand square is a coinduction square by previous slide.



- As on previous slide, obtain coinduction (z, g) where g is P -opcartesian.
- Obtain isomorphism h from universal property of coinductions.
- $Ph = \text{id}_J$ so h is opcartesian.
- f is opcartesian because g and h are.

Let $P: \mathbf{A} \rightarrow \mathbf{S}$ be a monoidal-extensive symmetric monoidal fibration.

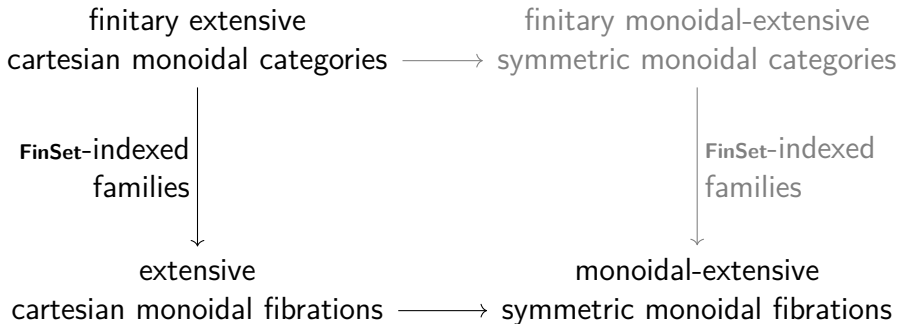
Conjecture

If P is a Beck-Chevalley bifibration with nice equalisers, then

- P lifts coinductions along P -opcartesian comonoid morphisms,*
- each commuting mixed square*

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ x \downarrow & & \downarrow y \\ A & \xrightarrow{\sigma_r A} & \Sigma_r A \end{array}$$

in $\mathbf{A}$ above a coinduction square in $\mathbf{S}$ is itself a coinduction square if and only if f is P -opcartesian.



Example (See Grunenfelder and Paré, 1987)

$(\mathbf{Vect}_{\mathbb{K}}, \times, 0)$ is not extensive, but $(\mathbf{Vect}_{\mathbb{K}}, \otimes_{\mathbb{K}}, \mathbb{K})$ is monoidal extensive.

Conjecture

Let $\mathcal{V}$ be a symmetric monoidal category with nice coreflexive equalisers. Then $\text{Comod}_{\mathcal{V}}$ is monoidal extensive.

- Introduced new notions of *monoidal extensivity* for indexed symmetric monoidal categories and symmetric monoidal fibrations
- Showed that they are equivalent.

Next time

- If $\mathcal{V}$ has nice coreflexive equalisers then the functor

$$\mathrm{Comod}_{\mathcal{V}}: \mathbf{Comod}_{\mathcal{V}} \rightarrow \mathbf{Comon}_{\mathcal{V}}$$

is a monoidal-extensive symmetric monoidal fibration with nice equalisers.

- Will show that the comodule fibration functor

$$\mathrm{Comod}_{(-)}: \mathbf{EqSymMonCat} \rightarrow \mathbf{EqExtSymMonFib}$$

is right adjoint to the underlying category functor.

m.dimeglio@ed.ac.uk